

gbawriter V4.1

Full controls · Halim Jarrar

Create and edit TXT files on your Game Boy Advance. Save your writing directly to the SD card. Put TXT files in **/gbawriter** at the root of your SD card, or import them from anywhere on the card (Start on Home). Requires a compatible Supercard SD.

Hardware-unverified build: emulator and host tests are not proof of safe saving on a physical Supercard. Back up your SD card and use disposable documents first.

Installation

- A GBA-compatible system and **Supercard SD** using the inherited SuperFW-compatible SD driver. Other flashcarts and generic emulator save files are not supported storage backends.
- Copy gbawriter.gba to your flashcart and launch it through its usual ROM loader. Published builds are on this repository's Releases page.
- Documents live in **/gbawriter/** on the SD card (created at start if missing). Settings and your list order live in the hidden folder /gbawriter/GBAWRITER.SYS; nothing is written next to your documents except the transient save files described below.
- Home needs the card, so it is read once at start (Checking SD...). Without a card or with an incompatible one, the driver takes a while to give up; Home then says No SD card, and Controls and Credits still work.
- Back up the SD card before testing. Never remove the card or power off while SAVING is displayed.

Home

Key	Action
Up / Down	Move (hold to scroll; a fresh press at either end wraps)
Left / Right	One page
A	New File: the date picker. A file: open it. The open file (▶): back into it. A acts when it is released.
Hold A + Up / Down	Move the selected file one row (Secret Settings > Reorder list, On by default). New File always stays first.
B	Back to the open file (B: Resume active file)
Start	Import
Select	Menu

At start the cursor is always on **New File**. The first time (no saved order), files are listed as before: diary dates newest first, then other names A to Z (upper/lower case ignored). After you move a file, that order is saved; files the order does not know yet (new, imported or copied on a PC) appear at the top, newest first. Rename keeps a file's place. Up to **1,000 files** are listed.

New File

The date picker proposes the day after the newest diary file in the chosen file name format (or after the last date you chose; at first **10 July 2026**). This is an editable preset, not a real-time clock. Up/Down choose the field, Left/Right change it, A creates, B goes back to Home. The fields follow the format (Month first for MM.DD.YYYY, Year first for YYYY-MM-DD); the file name is shown below. Invalid day combinations are clamped; years 1-9999.

Select > File names	Example
DDMMYYYY (default, as before V4.0)	01102026.txt
DD.MM.YYYY	01.10.2026.txt
MMDDYYYY	10012026.txt
MM.DD.YYYY	10.01.2026.txt
YYYY-MM-DD	2026-10-01.txt (also sorts by date on a PC)

The format only names new files; the list always shows the real names. DDMMYYYY and MMDDYYYY names cannot be told apart, so the proposal only counts files in the chosen format. **New File never overwrites an existing name**, including case-only .TXT differences: a collision shows FILE ALREADY EXISTS and returns to the date picker.

Leaving the editor and the open file

- **Start+A** saves (unchanged). **Start+B** goes to Home **without saving**; nothing is lost: the text stays in memory and the file is the open file: ► on its row and B: Resume active file at the bottom. B (or A on that row) returns to the text with the caret, view and Shift/Caps as they were.
- Opening another file or New File while the open file has unsaved changes asks **Unsaved changes: Save / Discard / Cancel** (B = Cancel). Save writes the file first and stops on a save error; Discard drops the changes when the other file opens.
- **There is no autosave: switching off loses text that is not saved.**

Import (Start on Home)

Import to /gbawriter starts at the SD-card root and lists folders (folder icon) and then TXT files, each A to Z. Hidden and system entries and /gbawriter itself are not shown. A opens a folder, B goes up, B at the root returns to Home. A on a TXT file asks Import into /gbawriter? (No first); Yes copies it with a percent display. When the name is used (also in other upper/lower case) the app offers Yes, as Name (2).txt (numbers 2 to 99). A failed copy leaves no partial file. Home then shows the new file at the top, selected, with Imported. The original stays where it is.

Select menu

On a file the title is its name and the first rows act on it:

- **Rename:** the name without .txt in a field, typed with the gbawriter letters below; Start+A saves, Start+B cancels; .txt is added. Not allowed: an empty name, / \ : * ? " < > |, a name ending in a space or a dot, a name already used (also in other upper/lower case). Changing only upper/lower case works.
- **Delete** (only when Secret Settings > Delete files is On): the row turns into Sure?; A again deletes the TXT file, B or moving cancels.
- The **open file** (►) cannot be renamed or deleted: This file is open. Open another file first. Files with recovery copies (.gwt, .gwb, .gwi) are refused with RECOVERY: CHECK SD ON PC.

Then for every row: **File names:** ... (A switches the format; the row shows the next proposed name), **Status bar: On / Off** (V4.1: whether the editor's bottom bar is shown when a file opens; saved, default On; Start+Select still toggles it while writing), the Controls topics (Files and menus, Letters, L layer, Spaces and case, Symbols, Accents, Caret and saving) and **Credits**. Pages: Left/Right turn, B returns to the menu.

Credits — page 1: gbawriter V4.1, Made by Halim Jarrar, (C) 2026, halimj.itch.io, gba@halim-jarrar.de; then License, Text and fonts, SD card and files, Engine, and **Secret Settings** (press A):

- Reorder list (A+DPAD): On (default On)
- Delete files: Off (default Off)
- Delete configuration: Sure?, A again removes GBAWRITER.SYS (settings and order back to their defaults; documents are never touched).

Typing

The editor and all typing controls are unchanged from V3.0, except that Start+B no longer saves (see above). The full-controls PDF has every control; regenerate it with python3 tools/build_controls_pdf.py (ReportLab and DejaVu Sans).

Hold a direction, then press **B / A / R** to choose its **first / second / third** letter. Lowercase is the default.

Direction	Normal: B / A / R	Hold L: B / A / R
Up	a / b / c	d / e / f
Right	h / i / j	k / l / m
Down	n / o / p	q / r / s
Left	t / u / w	x / y / z

L is held, never toggled. Diagonals are not letter groups.

- Keep Right held and press R twice: the first j becomes **g**.
- Keep Left held and press R twice: the first w becomes **v**.
- Release the group between presses to type literal jj or ww. These transformations depend on the same continuous hold, **not timing**, and never substitute unrelated text.
- **A alone:** space. **B alone:** backspace one UTF-8 codepoint; no effect at the beginning. Hold either button alone to repeat: one immediate edit, first repeat after **24 frames**, then every **5 frames**, exactly matching START navigation. Adding any other button cancels edit repeat; releasing a chord's modifier does not start it. Release A/B and press it alone again to rearm. Letter/SELECT/save chords remain press-only.
- **START released alone:** newline. Holding it does not insert immediately.

Shift and Caps

From normal case, **release a short isolated R press to arm one-shot Shift**. Hold **R alone continuously for 48 frames** (twice the initial A-repeat delay; about 0.8 seconds at 60 Hz) to enable **Caps while still held**, once per hold. Any other button before the threshold cancels that hold; releasing the companion cannot restart it. Release R and start a fresh solo hold.

With **Shift or Caps already active**, another isolated R press—short or long—**clears the mode on release**. It cannot arm Shift or Caps again during the same hold. Shift applies to the next successfully inserted alphabetic character; digits, spaces and punctuation do not consume it.

With Caps on, one isolated R press/release turns Caps off without arming Shift. R inside a direction group remains the third-letter key and cannot toggle Caps. Shift/Caps applies to the international letters below; ß stays ß, never two letters. Rejected capacity-limited edits do not consume Shift. A failed save preserves Shift/Caps as well as the document.

The bottom status bar reads **filename → active letter group → Shift/Caps**. The filename starts at x=8 (128-pixel slot), the group at x=144 (32-pixel slot), and Shift/Caps at x=184 (48-pixel slot): hold Up to see abc, or ABC with Shift/Caps; all normal and held-L groups follow the same rule. Release the direction to clear it; diagonals and START navigation show no letter group. * marks unsaved changes. Transient messages occupy only the filename slot, leaving capitalization and group visible.

START commands

Hold START, then:

Button	Action
Left / Right	Move caret one UTF-8 codepoint
Up / Down	Move to the closest horizontal position in the adjacent visual row
L / R	Previous / next viewport page, maintaining a valid caret
A	Save the current file
B	Back to Home without saving ; the text stays open (►, B: Resume active file)
SELECT	Toggle the bottom status bar / full-screen editor

Navigation repeats while held. Releasing START after a recognized command **does not add a newline**, even when saving fails. A save failure opens a clear error message and returns to the editor with text retained after acknowledgment. There is no autosave.

SELECT: one provisional character

Press SELECT alone to insert **.** **immediately**. Keep SELECT held to change **that same character**; release to keep it. Exception: SELECT after an eligible continuously held letter chord converts the existing letter instead (see International letters). Cycling never appends additional characters.

While SELECT is held	Cycle
Up	. → 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 0 → 1 ...
Down	. → 0 → 9 → 8 → 7 → 6 → 5 → 4 → 3 → 2 → 1 → 0 ...
R, without a direction	. → , → ' → " → : → ! → ? → .
L, without a completed letter chord	Exact reverse common-punctuation cycle
Right	. → (→) → / → ; → @ → # → % → & → _ → + → = → - → .
Left	Exact reverse additional-symbol cycle

A completed letter chord takes priority over intermediate directional/punctuation changes. For example SELECT, then Up, then L, then A changes the same provisional . to 1 and then to é, not three characters. A complete L-layer chord similarly overrides the temporary SELECT+L punctuation step. Unsupported accent families leave the provisional glyph unchanged. **START+SELECT toggles the status bar in the editor only**, in either press order. It cancels only a newly inserted SELECT provisional character, restoring the original caret and dirty state. A converted existing letter stays accented; it is not deleted. The chord never types, saves, or adds a newline. It toggles once per hold

and suppresses commands until both START and SELECT are released; other held buttons do not prevent rearming.

International letters

Accents work in either order: hold SELECT, then enter the normal letter combination; or enter a letter and keep its **exact single D-pad direction, L if used, and producing B/A/R button continuously held**, then press SELECT. That same letter changes to its first accented variant, keeping its case, without adding a period or another letter. There is no time limit, but releasing or changing the combination ends eligibility. Other extra buttons disqualify this letter-first gesture; diagonals are not letter groups. A letter with no accent variants stays unchanged in the letter-first order.

Example: hold L+UP and press A to type e; keep L+UP+A held and press SELECT to turn that same e into é. In the other order, hold SELECT, then L+UP and press A to produce é as one character. The held-L groups work in both orders too.

While SELECT and the same group remain held, **release and press the final B/A/R selector again** to cycle in the exact order below. Keep the direction and L layer (if used) held during cycling. Release SELECT to keep the current result. The letter-first result keeps the original letter's lowercase/Shift/Caps case, even after one-shot Shift was consumed.

Family	Lowercase cycle	Uppercase equivalents
A	á â ã ä å æ	Á Â Ã Ä Å Æ
C	ç ċ ċ	Ç Ć Ć
E	é è ê ë	É Ê Ë Ì
I	í î ï ð	Í Î Ï ð
N	ñ ñ	Ñ Ñ
O	ó ô õ ö ø œ	Ó Ô Õ Ö Ø Æ
S	ß š š	ß Š Š
U	ü ú û û	Ü Ú Û Ü
Y	ý ŷ	Ý Ÿ
Z	ž ž ž	Ž Ž Ž

E/S/Y/Z use the held L layer; N/O/U are unmodified groups. Cycles wrap back to the first alternate. All 77 distinct required lowercase/uppercase glyphs are verified against the real font packs and renderer, including even/odd pixel addressing. Home **SELECT** opens the menu with the Controls topics; Left/Right changes page; B returns. Help strings are checked for glyph coverage and screen width.

Storage and attribution

Files have no size limit: they stay on the SD card and only changes use RAM. Up to 64 KiB of typed text can be unsaved at once; when the bar shows BUFFER FULL - SAVE, save to continue (the text is kept). Saving a large file takes longer. Saves change only the opened TXT file. Settings and the list order are kept in the hidden folder /gbawriter/GBAWRITER.SYS. Temporary .gwt, .gwb and .gwi recovery files may remain after interrupted or failed saves. Do not delete them on the original card: back up the card and consult the README recovery instructions. There is no undo/redo or autosave: switching off loses text that is not saved.

Made by Halim Jarrar · (C) 2026 · halimj.itch.io · gba@halim-jarrar.de. gbawriter is free software under the GNU GPL 3.0 or later. Text renderer: SuperFW by David Guillen Fandos (GPL 3.0 or later). Fonts:

UNSCII by Viznut (GPL) and GNU Unifont, also its Hangul blocks (GPL 2.0 or later). UI font: from gbamp3 (MIT). SD card driver: SuperFW (GPL 3.0 or later). File system: FatFs by ChaN (BSD style). Engine: Butano by Gustavo Valiente (zlib); built with devkitPro / devkitARM. Font and vendor notices remain in the source; see LICENSE and README.